Results for XES3G5M-sub-small

Original held-out split; N = 5, 10, ..., 50.

Metrics: Adaptability (Ada), Novelty (NOV), and Ep at thresholds 0.4, 0.5, 0.6, 0.7.

Cell format: mean +/- sample SD; deterministic CF baselines show one value.

Bold = best; underline = second best by unrounded mean.

Ablation methods
2CKG4ER
No-mastery
No-forgetting
No-student-feature
No-exercise-feature
ID-based only

Comparison methods
EB-CF
SB-CF
CBF
TransE
TransE-adv
RotatE
DistMult
ComplEx
2CKG4ER

Experimental setting

All methods use the selected drop-zero recommendation graph for this dataset.

Graph parameters: theta_m = 0.8, theta_f = 0.8, rec = 50.

Source: Data_Fin_New experiment metrics; all methods included.

Ablation performance

Six variants evaluated on the selected graph.

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.879 ± 0.003**	0.802 ± 0.003	0.878 ± 0.019	0.853 ± 0.006	0.872 ± 0.002	0.862 ± 0.011
10	0.876 ± 0.003	0.806 ± 0.011	**0.880 ± 0.010**	0.853 ± 0.004	0.872 ± 0.003	0.861 ± 0.009
15	0.873 ± 0.004	0.812 ± 0.002	**0.878 ± 0.009**	0.852 ± 0.004	0.871 ± 0.003	0.860 ± 0.008
20	0.873 ± 0.003	0.818 ± 0.006	**0.877 ± 0.009**	0.852 ± 0.004	0.871 ± 0.003	0.858 ± 0.007
25	0.873 ± 0.003	0.825 ± 0.004	**0.877 ± 0.009**	0.851 ± 0.004	0.871 ± 0.002	0.858 ± 0.006
30	0.872 ± 0.002	0.829 ± 0.002	**0.877 ± 0.008**	0.851 ± 0.003	0.871 ± 0.002	0.857 ± 0.005
35	0.871 ± 0.002	0.832 ± 0.001	**0.877 ± 0.008**	0.851 ± 0.003	0.870 ± 0.002	0.857 ± 0.004
40	0.871 ± 0.002	0.837 ± 0.001	**0.876 ± 0.007**	0.851 ± 0.003	0.870 ± 0.002	0.856 ± 0.004
45	0.870 ± 0.003	0.842 ± 0.000	**0.876 ± 0.006**	0.851 ± 0.003	0.869 ± 0.002	0.856 ± 0.004
50	0.870 ± 0.003	0.845 ± 0.000	**0.876 ± 0.005**	0.850 ± 0.003	0.869 ± 0.002	0.855 ± 0.003

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**1.000 ± 0.000**	1.000 ± 0.000	0.998 ± 0.001	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
10	**1.000 ± 0.000**	1.000 ± 0.000	0.997 ± 0.001	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
15	1.000 ± 0.000	**1.000 ± 0.000**	0.997 ± 0.002	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
20	1.000 ± 0.000	**1.000 ± 0.000**	0.996 ± 0.002	0.999 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
25	1.000 ± 0.000	**1.000 ± 0.000**	0.996 ± 0.002	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
30	1.000 ± 0.000	**1.000 ± 0.000**	0.994 ± 0.002	0.999 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
35	1.000 ± 0.000	**1.000 ± 0.000**	0.994 ± 0.001	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
40	1.000 ± 0.000	**1.000 ± 0.000**	0.993 ± 0.002	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
45	1.000 ± 0.000	**1.000 ± 0.000**	0.992 ± 0.003	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
50	1.000 ± 0.000	1.000 ± 0.000	0.992 ± 0.003	1.000 ± 0.000	1.000 ± 0.000	**1.000 ± 0.000**

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Ablation performance

Six variants evaluated on the selected graph.

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.143 ± 0.045	**0.243 ± 0.018**	0.043 ± 0.059	0.194 ± 0.016	0.158 ± 0.038	0.212 ± 0.033
10	0.234 ± 0.065	0.249 ± 0.128	0.032 ± 0.099	0.275 ± 0.018	0.264 ± 0.014	**0.285 ± 0.031**
15	0.297 ± 0.042	0.286 ± 0.034	0.034 ± 0.098	0.327 ± 0.009	0.299 ± 0.010	**0.330 ± 0.027**
20	0.338 ± 0.022	0.339 ± 0.052	0.035 ± 0.118	0.347 ± 0.011	**0.355 ± 0.024**	0.353 ± 0.021
25	0.353 ± 0.030	**0.436 ± 0.007**	0.056 ± 0.136	0.361 ± 0.004	0.339 ± 0.034	0.369 ± 0.009
30	0.363 ± 0.017	**0.452 ± 0.043**	0.104 ± 0.116	0.374 ± 0.003	0.347 ± 0.010	0.379 ± 0.009
35	0.378 ± 0.033	**0.513 ± 0.029**	0.154 ± 0.077	0.376 ± 0.005	0.374 ± 0.015	0.384 ± 0.003
40	0.378 ± 0.017	**0.541 ± 0.023**	0.206 ± 0.044	0.378 ± 0.005	0.379 ± 0.015	0.388 ± 0.005
45	0.394 ± 0.012	**0.500 ± 0.045**	0.214 ± 0.045	0.378 ± 0.001	0.389 ± 0.038	0.383 ± 0.004
50	0.391 ± 0.007	**0.465 ± 0.075**	0.237 ± 0.042	0.372 ± 0.005	0.379 ± 0.027	0.386 ± 0.009

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.138 ± 0.045	**0.238 ± 0.020**	0.040 ± 0.057	0.189 ± 0.015	0.154 ± 0.039	0.208 ± 0.033
10	0.230 ± 0.064	0.246 ± 0.128	0.021 ± 0.090	0.271 ± 0.018	0.258 ± 0.017	**0.283 ± 0.032**
15	0.292 ± 0.040	0.283 ± 0.034	0.024 ± 0.092	0.322 ± 0.010	0.294 ± 0.011	**0.327 ± 0.028**
20	0.333 ± 0.020	0.336 ± 0.053	0.022 ± 0.108	0.342 ± 0.011	**0.351 ± 0.023**	0.351 ± 0.022
25	0.349 ± 0.028	**0.434 ± 0.008**	0.043 ± 0.128	0.356 ± 0.003	0.336 ± 0.034	0.367 ± 0.010
30	0.359 ± 0.015	**0.450 ± 0.043**	0.093 ± 0.112	0.370 ± 0.004	0.344 ± 0.010	0.377 ± 0.009
35	0.374 ± 0.032	**0.513 ± 0.029**	0.134 ± 0.072	0.372 ± 0.005	0.371 ± 0.015	0.383 ± 0.002
40	0.375 ± 0.014	**0.541 ± 0.023**	0.186 ± 0.041	0.375 ± 0.005	0.377 ± 0.015	0.387 ± 0.005
45	0.391 ± 0.009	**0.500 ± 0.045**	0.195 ± 0.038	0.376 ± 0.002	0.387 ± 0.037	0.382 ± 0.004
50	0.389 ± 0.004	**0.465 ± 0.075**	0.216 ± 0.035	0.370 ± 0.004	0.378 ± 0.026	0.386 ± 0.009

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Ablation performance

Six variants evaluated on the selected graph.

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.127 ± 0.043	**0.229 ± 0.022**	0.037 ± 0.057	0.178 ± 0.013	0.137 ± 0.040	0.200 ± 0.030
10	0.216 ± 0.061	0.238 ± 0.127	0.005 ± 0.078	0.256 ± 0.020	0.232 ± 0.025	**0.272 ± 0.032**
15	0.277 ± 0.033	0.277 ± 0.035	0.001 ± 0.079	0.306 ± 0.012	0.268 ± 0.016	**0.315 ± 0.032**
20	0.317 ± 0.013	0.331 ± 0.053	-0.014 ± 0.101	0.324 ± 0.014	0.324 ± 0.025	**0.340 ± 0.028**
25	0.332 ± 0.020	**0.429 ± 0.008**	0.003 ± 0.122	0.339 ± 0.004	0.309 ± 0.039	0.357 ± 0.017
30	0.342 ± 0.011	**0.447 ± 0.043**	0.049 ± 0.112	0.354 ± 0.004	0.317 ± 0.015	0.367 ± 0.016
35	0.359 ± 0.026	**0.511 ± 0.029**	0.092 ± 0.076	0.357 ± 0.003	0.346 ± 0.014	0.375 ± 0.006
40	0.359 ± 0.006	**0.540 ± 0.023**	0.133 ± 0.047	0.360 ± 0.008	0.354 ± 0.016	0.380 ± 0.003
45	0.376 ± 0.008	**0.500 ± 0.045**	0.141 ± 0.032	0.362 ± 0.005	0.365 ± 0.030	0.375 ± 0.008
50	0.374 ± 0.008	**0.465 ± 0.075**	0.155 ± 0.034	0.358 ± 0.001	0.357 ± 0.018	0.380 ± 0.004

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.091 ± 0.045	**0.211 ± 0.025**	0.033 ± 0.056	0.157 ± 0.010	0.100 ± 0.047	0.177 ± 0.029
10	0.164 ± 0.061	0.218 ± 0.120	-0.024 ± 0.059	0.223 ± 0.020	0.165 ± 0.040	**0.241 ± 0.031**
15	0.208 ± 0.023	0.256 ± 0.030	-0.033 ± 0.055	0.265 ± 0.014	0.192 ± 0.026	**0.282 ± 0.033**
20	0.243 ± 0.006	**0.307 ± 0.048**	-0.063 ± 0.082	0.277 ± 0.021	0.241 ± 0.040	0.304 ± 0.032
25	0.251 ± 0.011	**0.399 ± 0.010**	-0.059 ± 0.098	0.289 ± 0.016	0.215 ± 0.057	0.317 ± 0.026
30	0.262 ± 0.010	**0.417 ± 0.044**	-0.028 ± 0.095	0.301 ± 0.014	0.221 ± 0.031	0.325 ± 0.026
35	0.277 ± 0.019	**0.481 ± 0.031**	0.005 ± 0.070	0.304 ± 0.019	0.247 ± 0.017	0.331 ± 0.020
40	0.276 ± 0.005	**0.515 ± 0.023**	0.031 ± 0.042	0.308 ± 0.016	0.256 ± 0.019	0.335 ± 0.020
45	0.293 ± 0.011	**0.480 ± 0.045**	0.025 ± 0.056	0.309 ± 0.020	0.267 ± 0.017	0.331 ± 0.025
50	0.291 ± 0.019	**0.448 ± 0.076**	0.027 ± 0.055	0.303 ± 0.024	0.258 ± 0.020	0.334 ± 0.019

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(a) Adaptability (Ada)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.821	0.834	0.549	0.844 ± 0.003	0.863 ± 0.004	0.789 ± 0.010	0.862 ± 0.003	0.856 ± 0.004	**0.879 ± 0.003**
10	0.811	0.840	0.577	0.849 ± 0.002	0.862 ± 0.002	0.816 ± 0.006	0.861 ± 0.002	0.855 ± 0.003	**0.876 ± 0.003**
15	0.811	0.844	0.594	0.850 ± 0.001	0.860 ± 0.001	0.827 ± 0.006	0.859 ± 0.002	0.854 ± 0.001	**0.873 ± 0.004**
20	0.813	0.846	0.604	0.850 ± 0.001	0.858 ± 0.003	0.833 ± 0.006	0.857 ± 0.002	0.852 ± 0.001	**0.873 ± 0.003**
25	0.815	0.846	0.613	0.848 ± 0.002	0.855 ± 0.002	0.836 ± 0.006	0.857 ± 0.002	0.850 ± 0.000	**0.873 ± 0.003**
30	0.817	0.847	0.621	0.846 ± 0.002	0.853 ± 0.001	0.837 ± 0.005	0.855 ± 0.003	0.848 ± 0.001	**0.872 ± 0.002**
35	0.818	0.845	0.628	0.845 ± 0.002	0.852 ± 0.001	0.838 ± 0.005	0.855 ± 0.003	0.847 ± 0.002	**0.871 ± 0.002**
40	0.820	0.842	0.633	0.846 ± 0.001	0.851 ± 0.001	0.839 ± 0.005	0.854 ± 0.004	0.846 ± 0.001	**0.871 ± 0.002**
45	0.821	0.840	0.638	0.845 ± 0.001	0.851 ± 0.001	0.839 ± 0.004	0.854 ± 0.004	0.846 ± 0.002	**0.870 ± 0.003**
50	0.822	0.839	0.642	0.846 ± 0.001	0.851 ± 0.001	0.838 ± 0.004	0.853 ± 0.004	0.845 ± 0.002	**0.870 ± 0.003**

(b) Novelty (NOV)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.998	1.000	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
10	0.998	1.000	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000
15	0.999	1.000	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000
20	0.999	0.999	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
25	0.998	0.999	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
30	0.996	0.998	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
35	0.994	0.997	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
40	0.993	0.998	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
45	0.993	0.998	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
50	0.992	0.998	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(c) Ep, threshold 0.4

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.021	0.182	0.117	0.199 ± 0.017	**0.213 ± 0.023**	0.169 ± 0.009	0.170 ± 0.028	0.169 ± 0.013	0.143 ± 0.045
10	-0.023	0.182	0.191	0.300 ± 0.017	**0.331 ± 0.017**	0.246 ± 0.011	0.232 ± 0.026	0.234 ± 0.005	0.234 ± 0.065
15	-0.054	0.239	0.244	0.352 ± 0.006	**0.370 ± 0.024**	0.302 ± 0.020	0.271 ± 0.016	0.275 ± 0.008	0.297 ± 0.042
20	-0.051	0.210	0.284	0.384 ± 0.005	**0.399 ± 0.018**	0.342 ± 0.014	0.289 ± 0.022	0.305 ± 0.007	0.338 ± 0.022
25	-0.029	0.202	0.293	0.380 ± 0.004	**0.394 ± 0.010**	0.350 ± 0.010	0.299 ± 0.017	0.320 ± 0.004	0.353 ± 0.030
30	-0.024	0.206	0.308	0.379 ± 0.004	**0.393 ± 0.002**	0.360 ± 0.002	0.304 ± 0.022	0.316 ± 0.015	0.363 ± 0.017
35	-0.049	0.192	0.311	0.375 ± 0.006	**0.385 ± 0.002**	0.365 ± 0.003	0.312 ± 0.019	0.321 ± 0.005	0.378 ± 0.033
40	-0.050	0.196	0.312	0.372 ± 0.006	**0.380 ± 0.003**	0.364 ± 0.004	0.317 ± 0.011	0.325 ± 0.015	0.378 ± 0.017
45	-0.055	0.218	0.318	0.366 ± 0.005	0.373 ± 0.001	0.361 ± 0.004	0.316 ± 0.008	0.331 ± 0.009	**0.394 ± 0.012**
50	-0.038	0.253	0.325	0.363 ± 0.009	0.374 ± 0.003	0.364 ± 0.004	0.316 ± 0.004	0.331 ± 0.002	**0.391 ± 0.007**

(d) Ep, threshold 0.5

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.021	0.175	0.079	0.196 ± 0.016	**0.206 ± 0.025**	0.167 ± 0.009	0.162 ± 0.029	0.159 ± 0.014	0.138 ± 0.045
10	-0.023	0.174	0.151	0.296 ± 0.017	**0.324 ± 0.018**	0.242 ± 0.012	0.225 ± 0.028	0.224 ± 0.007	0.230 ± 0.064
15	-0.054	0.232	0.208	0.348 ± 0.006	**0.364 ± 0.028**	0.299 ± 0.019	0.264 ± 0.018	0.267 ± 0.008	0.292 ± 0.040
20	-0.052	0.201	0.254	0.380 ± 0.003	**0.394 ± 0.021**	0.339 ± 0.011	0.282 ± 0.022	0.300 ± 0.007	0.333 ± 0.020
25	-0.030	0.193	0.268	0.377 ± 0.006	**0.390 ± 0.013**	0.347 ± 0.009	0.293 ± 0.018	0.315 ± 0.007	0.349 ± 0.028
30	-0.024	0.195	0.291	0.377 ± 0.003	**0.390 ± 0.004**	0.358 ± 0.001	0.299 ± 0.023	0.312 ± 0.014	0.359 ± 0.015
35	-0.049	0.181	0.295	0.373 ± 0.005	**0.383 ± 0.004**	0.364 ± 0.002	0.307 ± 0.019	0.318 ± 0.006	0.374 ± 0.032
40	-0.051	0.186	0.299	0.371 ± 0.005	**0.378 ± 0.001**	0.363 ± 0.005	0.313 ± 0.011	0.323 ± 0.015	0.375 ± 0.014
45	-0.055	0.210	0.308	0.366 ± 0.005	0.371 ± 0.002	0.360 ± 0.004	0.312 ± 0.009	0.330 ± 0.009	**0.391 ± 0.009**
50	-0.038	0.247	0.320	0.363 ± 0.009	0.373 ± 0.003	0.364 ± 0.005	0.312 ± 0.006	0.330 ± 0.002	**0.389 ± 0.004**

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(e) Ep, threshold 0.6

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.021	0.166	0.050	0.184 ± 0.017	**0.192 ± 0.024**	0.155 ± 0.008	0.150 ± 0.028	0.150 ± 0.011	0.127 ± 0.043
10	-0.023	0.163	0.114	0.281 ± 0.018	**0.311 ± 0.020**	0.225 ± 0.013	0.212 ± 0.024	0.211 ± 0.006	0.216 ± 0.061
15	-0.054	0.220	0.159	0.331 ± 0.009	**0.350 ± 0.029**	0.281 ± 0.022	0.253 ± 0.014	0.250 ± 0.008	0.277 ± 0.033
20	-0.052	0.186	0.200	0.363 ± 0.003	**0.378 ± 0.022**	0.319 ± 0.015	0.271 ± 0.020	0.282 ± 0.009	0.317 ± 0.013
25	-0.030	0.171	0.219	0.361 ± 0.005	**0.374 ± 0.014**	0.327 ± 0.011	0.282 ± 0.013	0.297 ± 0.008	0.332 ± 0.020
30	-0.024	0.165	0.246	0.361 ± 0.007	**0.374 ± 0.005**	0.339 ± 0.002	0.287 ± 0.017	0.296 ± 0.016	0.342 ± 0.011
35	-0.049	0.147	0.254	0.359 ± 0.010	**0.368 ± 0.004**	0.345 ± 0.003	0.293 ± 0.014	0.304 ± 0.009	0.359 ± 0.026
40	-0.051	0.151	0.254	0.357 ± 0.008	**0.364 ± 0.002**	0.345 ± 0.007	0.300 ± 0.007	0.309 ± 0.016	0.359 ± 0.006
45	-0.055	0.178	0.270	0.353 ± 0.001	0.358 ± 0.002	0.344 ± 0.003	0.300 ± 0.006	0.318 ± 0.012	**0.376 ± 0.008**
50	-0.038	0.216	0.284	0.351 ± 0.006	0.360 ± 0.002	0.349 ± 0.006	0.301 ± 0.007	0.319 ± 0.002	**0.374 ± 0.008**

(f) Ep, threshold 0.7

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.022	0.132	-0.010	**0.167 ± 0.013**	0.157 ± 0.030	0.132 ± 0.007	0.126 ± 0.026	0.123 ± 0.009	0.091 ± 0.045
10	-0.024	0.127	0.032	0.260 ± 0.018	**0.269 ± 0.031**	0.192 ± 0.010	0.185 ± 0.022	0.174 ± 0.011	0.164 ± 0.061
15	-0.055	0.184	0.070	**0.306 ± 0.011**	0.303 ± 0.038	0.240 ± 0.020	0.216 ± 0.011	0.208 ± 0.012	0.208 ± 0.023
20	-0.053	0.141	0.105	**0.335 ± 0.006**	0.330 ± 0.033	0.273 ± 0.019	0.229 ± 0.020	0.235 ± 0.014	0.243 ± 0.006
25	-0.031	0.117	0.106	**0.330 ± 0.007**	0.325 ± 0.027	0.280 ± 0.013	0.235 ± 0.016	0.250 ± 0.005	0.251 ± 0.011
30	-0.025	0.093	0.127	**0.330 ± 0.012**	0.324 ± 0.024	0.290 ± 0.010	0.238 ± 0.016	0.248 ± 0.023	0.262 ± 0.010
35	-0.053	0.041	0.132	**0.327 ± 0.016**	0.320 ± 0.023	0.295 ± 0.012	0.241 ± 0.014	0.254 ± 0.008	0.277 ± 0.019
40	-0.056	0.048	0.131	**0.325 ± 0.015**	0.318 ± 0.025	0.295 ± 0.013	0.244 ± 0.008	0.259 ± 0.020	0.276 ± 0.005
45	-0.063	0.054	0.150	**0.320 ± 0.007**	0.311 ± 0.025	0.291 ± 0.008	0.244 ± 0.011	0.268 ± 0.009	0.293 ± 0.011
50	-0.045	0.099	0.177	**0.318 ± 0.002**	0.314 ± 0.026	0.296 ± 0.009	0.241 ± 0.003	0.268 ± 0.006	0.291 ± 0.019

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.